

SIMATS ENGINEERING

ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5103

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

Annexure A

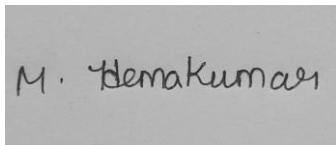
APPLICATION- BASED QUESTIONS

Code of Conduct:

I [M.Hemakumar(192521138)] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in black ink. The signature appears to be "M. Hemakumar" written in a cursive, slightly slanted script.

Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Deduce the most likely original message.

Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CSA5103 – CRYPTOGRAPHY AND NETWORK SECURITY

ASSESSMENT TOOL 1 – APPLICATION-BASED QUESTIONS

CO1: Apply classical encryption techniques using symmetric/asymmetric ciphers to encrypt/decrypt data for data security.

CSA5103 – CRYPTOGRAPHY AND NETWORK SECURITY

QUESTION 3 – Vigenère Cipher

=Given: Plaintext = HELLO; Keyword = KEY

1. Introduction

The Vigenère Cipher is a classical polyalphabetic substitution cipher used to encrypt messages. Unlike the Caesar Cipher, which uses a single fixed shift for every letter, the Vigenère Cipher uses different shifts based on the letters of a keyword. The keyword is repeated until it matches the length of the plaintext.

2. Encryption Principle

The letters are represented using numerical values: A = 0, B = 1, C = 2, ..., Z = 25.

Encryption formula: $C = (P + K) \bmod 26$

Where C = Ciphertext value, P = Plaintext value, and K = Keyword value.

3. Repeating the Keyword

Plaintext: H E L L O

Keyword: K E Y K E

The keyword KEY is repeated as KEYKE to match the five letters of the plaintext.

4. Step-by-Step Encryption

Plaintext	Value	Key	Key Value	Calculation	Cipher
H	7	K	10	$(7+10) \bmod 26 = 17$	R
E	4	E	4	$(4+4) \bmod 26 = 8$	I
L	11	Y	24	$(11+24) \bmod 26 = 9$	J
L	11	K	10	$(11+10) \bmod 26 = 21$	V
O	14	E	4	$(14+4) \bmod 26 = 18$	S

Therefore: Plaintext = HELLO; Keyword = KEY; Ciphertext = RIJVS.

5. How the Keyword Influences Encryption

The keyword KEY determines the shift applied to each character. K shifts H by 10 positions to R; E shifts E by 4 positions to I; Y shifts L by 24 positions to J; the keyword then repeats, so K shifts L to V and E shifts O to S. Thus, the same plaintext letter can potentially produce different ciphertext letters depending on which keyword letter is aligned with it.

6. Decryption

For decryption, the process is reversed using: $P = (C - K) \bmod 26$. The receiver uses the same keyword KEY to recover the original plaintext.

Ciphertext = RIJVS; Keyword = KEY; Plaintext = HELLO.

7. Security Analysis

The Vigenère Cipher is stronger than a single-shift Caesar Cipher because it uses multiple substitution alphabets rather than one fixed substitution. However, a short repeating keyword can create patterns that may be exploited through cryptanalysis.

8. Advantages

- It is more difficult to break than a simple Caesar Cipher.
- It uses multiple substitution alphabets.
- A keyword controls the encryption process.
- The same plaintext letter does not necessarily produce the same ciphertext letter.

9. Limitations

- A short or repeated keyword can create detectable patterns.
- It is vulnerable to classical cryptanalysis techniques.
- It is not suitable for protecting sensitive information in modern computer networks.

10. Conclusion

The Vigenère Cipher encrypts HELLO using the keyword KEY by repeating the keyword as KEYKE.

Applying the corresponding shifts produces the ciphertext RIJVS. It provides better security than a simple Caesar Cipher but is still considered a classical encryption method and should not be used as a modern secure encryption algorithm.

QUESTION 4 – Image Steganography

1. Introduction

Image Steganography is a technique used to hide secret information inside an ordinary image. The image appears normal to a person viewing it, while confidential information is concealed within the image data. A common technique is Least Significant Bit (LSB) steganography, in which small changes are made to the least significant bits of image pixels.

2. Principle of Image Steganography

A digital image is made up of pixels. Each pixel contains numerical values representing its color. For example, suppose a pixel contains 11001010. If the secret bit is 1, the last bit can be changed: 11001010 → 11001011.

This very small change usually has little or no visible effect on the image. Therefore, secret information can be embedded into the image without making an obvious visual change.

3. LSB Steganography

LSB stands for Least Significant Bit. In this method, the least significant bit of a pixel value is modified to store secret information.

Original Pixel	Secret Bit	Modified Pixel
10110100	1	10110101
11001011	0	11001010
11110010	1	11110011
10011001	0	10011000

4. How Secret Data is Hidden

=The basic process is: Secret Message → Convert to Binary → Select Image Pixels → Modify LSBs → Stego Image.

For example, if the secret message is converted into 10110010, the bits can be embedded into the LSBs of eight selected pixel values. The resulting image is called a stego image because it contains hidden information.

5. Detection / Investigation

a. LSB Analysis

Investigators examine the least significant bits of image pixels. Unusual or non-random patterns may indicate that data has been embedded.

b. Statistical Analysis

Statistical techniques can be used to examine pixel distributions and identify abnormalities that may result from hidden data.

c. Metadata Examination

Investigators can inspect file metadata, file size, image format, file structure, and unusual metadata fields. Unexpected information may provide clues about hidden content.

d. Comparison with Original Image

If an original version of the image is available, investigators can compare it with the suspicious image. Differences in pixel values or file properties can help identify possible data embedding.

6. Techniques to Strengthen Hidden Communication

Technique 1 – Encrypt Before Embedding

The secret message should first be encrypted and then embedded into the image. Message → Encryption → Ciphertext → Steganography → Image. Even if an attacker extracts the hidden data, the extracted information will still be encrypted.

Technique 2 – Key-Based / Adaptive Embedding

Instead of always hiding data in predictable pixel locations, a secret key can determine where the information is embedded. This makes detection and extraction more difficult because the attacker does not know the embedding locations.

7. Advantages

- The image can appear normal to casual viewers.
- It provides concealment of communication.
- LSB techniques are relatively simple to implement.
- It can be combined with encryption for additional protection.

8. Limitations

- Hidden data may be detected through statistical analysis.
- Image compression or modification may destroy the hidden information.
- Steganography alone does not provide strong confidentiality.
- If hidden data is extracted, it may be readable unless it was encrypted first.

9. Security Analysis

Steganography mainly provides concealment, rather than strong confidentiality. Therefore, hiding a message alone is not sufficient for modern security requirements. A stronger approach is Encryption + Steganography. First encrypt the confidential message, then hide the encrypted message inside the image. This provides an additional layer of security.

10. Conclusion

Image steganography hides confidential information inside an ordinary image. LSB steganography is a common approach in which small changes are made to pixel bits. Investigators can detect possible hidden information through LSB analysis, statistical analysis, metadata examination, and comparison with an original image. For stronger security, the secret message should be encrypted before embedding, and key-based or adaptive embedding can be used.